

Basics of Programming and Algorithm

C++ | Java | C# | Python



Byte By Byte CS
AFG. Future Technology
Mortaza Nabavi | Spring 2025

About the Course

Course Tittle

Basics of Algorithm and Programming

Course Tittle

CS-SE0107 / CS-IT0107

Course Code

Computer Science Faculty, Gharjistan University

Organizer

Tuesdays, 05:00 AM @ PC-Pool

Time & Location

Mortaza Nabavi

Instructor

Spring 2025 (1405-1)

Semester



About the Course

Outlines & Tittles

Module 0: Why Computer & Programming?

Computer | Computer Types | Programming Languages

Module 1: Planning

Problem-Solving | Algorithm | Flow Chart | Sodo Code

Module 2: Introduction to C++

History | IDE | Setup & Initializing | Simple Commands

Module 3: Basic Programs

Input | Output | Data Types | Assignments | Calculations

Module 4: Conditional Statement

Condition in C++

Module 5: Loops & Iteration

Manual Loop | Loop in C++



About the Course

Addresses

lms.futech.af

the LMS

@ClassBuzz

Telegram Channel, Class Updates and Announcements

mortaza@ghu.edu.af &

MortazaNabavi@outlook.com &

@MortazaNabavi

My Inboxes & Telegram



About the Course

Grading Breakdown

Grading Leagues

	Final Exam	Mid. Exam	Assignments	Attendance	Total
League #1	60	20	20	15	115
League #2	60	20	15	-	95
League #3	60	20	-	-	80
League #4	70	-	-	-	70

* You'll receive the Max. among the Leagues ;)

Assignments Coefficients

On-Time: 1.2

1-Day Delay: 0.8

Semester Delay: 0.5

* You'll receive half of the assignment's point by just submitting ;)



Why Computer & Programming?

Module 0



What is COMPUTER ?



M0: Why Computer & Programming?

What is Computer?

Computer Definition

Computer is an electronic device that **processes** data to perform tasks, **following instructions**.

Processing Data

Input, process, output, storage

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?

Speed Computers perform billions of operations per second

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful? { Speed

Accuracy Computers execute tasks without errors if programmed correctly

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?

{
Speed
Accuracy

Storage Can store vast amounts of data reliably (from gigabytes to petabytes)

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?

Speed

Accuracy

Storage

Versatility

Adaptable to countless tasks, from word processing to scientific simulations

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?

Speed

Accuracy

Storage

Versatility

Automation

Handles repetitive tasks efficiently without intervention

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?

Speed
Accuracy
Storage
Versatility
Automation

Diligence

Works continuously without getting tired or bored

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?

Speed
Accuracy
Storage
Versatility
Automation
Diligence

Communication Can communicate and share resources with others

M0: Why Computer & Programming?

Key Properties of Computers

What makes computers powerful?



- Speed
- Accuracy
- Storage
- Versatility
- Automation
- Diligence
- Communication

**Can you Name
some types of
COMPUTERs**



M0: Why Computer & Programming?

Computers Are Every Where!



These are Computers;

But, Laptops & Desktops are **NOT** ALL Computers!

M0: Why Computer & Programming?

Key Properties of Computers

Some Type of Computers



Personal Computers

Everyday devices for individual use, including desktops, laptops, and tablets.



M0: Why Computer & Programming?

Key Properties of Computers

Some Type of Computers

Personal Computers

Servers

Powerful machines that manage networks, store data, and handle requests from multiple users.



M0: Why Computer & Programming?

Key Properties of Computers

Some Type of Computers

Personal Computers

Servers

Supercomputers

Massive systems for complex computations, often used in research and simulations.



M0: Why Computer & Programming?

Key Properties of Computers

Some Type of Computers

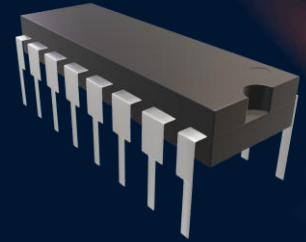
Personal Computers

Servers

Supercomputers

Embedded Systems

Specialized computers integrated into other devices, with limited resources but specific tasks.



M0: Why Computer & Programming?

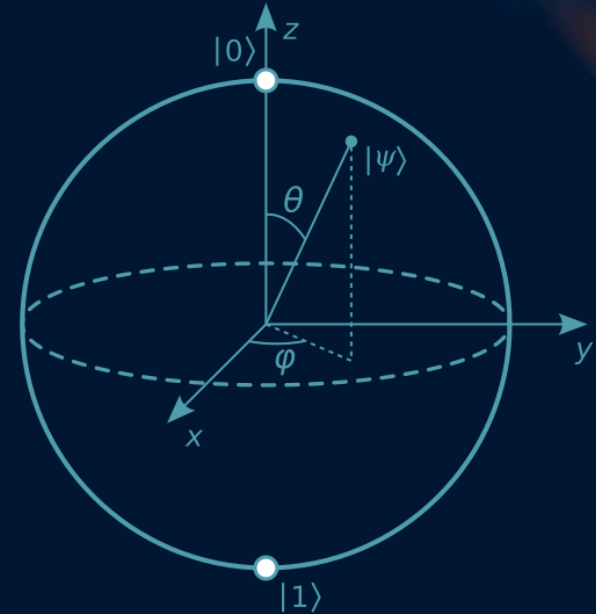
Key Properties of Computers

Some Type of Computers

Personal Computers
Servers
Supercomputers
Embedded Systems

Quantum Computers

Emerging tech using quantum bits (qubits) for solving problems impossible for classical computers.



M0: Why Computer & Programming?

Key Properties of Computers

Some Type of Computers

- Personal Computers
- Servers
- Supercomputers
- Embedded Systems
- Quantum Computers

M0: Why Computer & Programming?

Computers by Purpose

Is this a **COMPUTER**?

Name its properties!



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M0: Why Computer & Programming?

Computers by Purpose



Specific Purpose Use



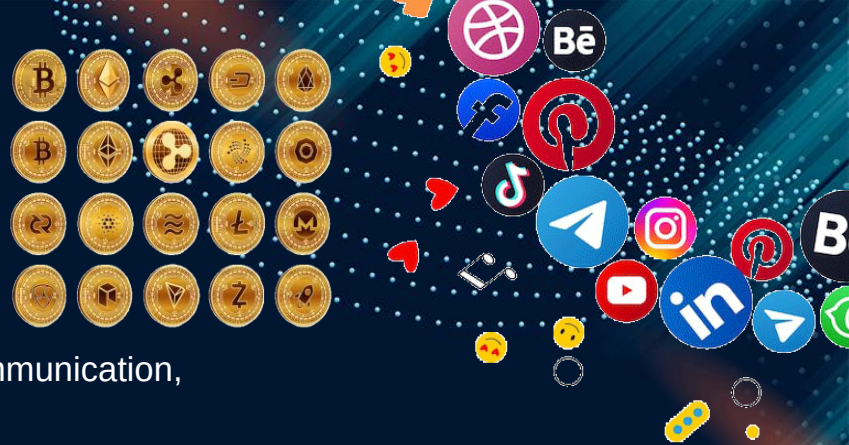
General Purpose Use

M0: Why Computer & Programming?

The Importance of Computers in Modern Life

Everyday Impact

Computers power Education, Healthcare, Entertainment, Communication, Economy & Business, Industry and ...



M0: Why Computer & Programming?

The Importance of Computers in Modern Life

Everyday Impact

Societal and Economic Role

Job Creation, Innovation, Space Exploration,
Global Connectivity



M0: Why Computer & Programming?

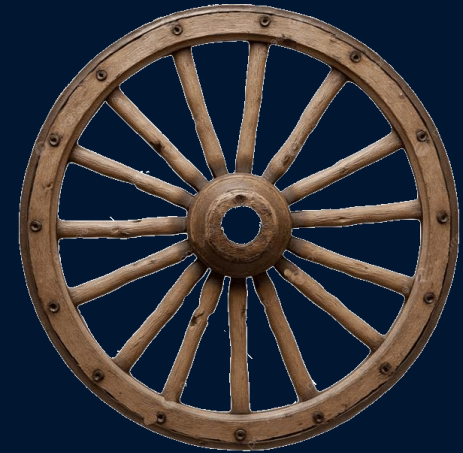
The Importance of Computers in Modern Life

Everyday Impact

Societal and Economic Role

Future Relevance

Computers aren't just tools; they're the foundation of tomorrow's inventions.



What is PROGRAMMING



M0: Why Computer & Programming?

What is Programming?

Programming Definition

The process of writing instructions (code) for computers to solve problems or perform tasks

Programming Process

Problem-Solving, Planning (Algorithms), Coding, Testing and Debugging

Programming Languages

The instructions are being written by programming languages

What is Natural Language



M0: Why Computer & Programming?

Natural Language

Definition

The languages which we (humans) use to communicate ☺





What is Machine Code

M0: Why Computer & Programming?

Machine Code

Definition

The code (0-1) which digital systems process



M0: Why Computer & Programming?

High & Low Level Programming Languages

Programming
languages

High-Level

Java, C++, C#, Python

Low-Level

Assembly

Natural languages



Machine Code